

Referee Report: “BBP Spectral Proxy: A Computable Alternative to SCX Theorem 2’s Mutual Information”

Review requested by: Editor-in-Chief (hostile review mode) **Venue:** Annals of Statistics (imagined) **Recommendation:** REJECT

Executive Summary

This document attempts to bridge two mathematical frameworks: (1) the BBP phase transition for spiked covariance matrices, and (2) the SCX Theorem 2 bound connecting mutual information $I(\phi; S)$ to F1 detection performance. The goal is admirable — replace an intractable quantity (mutual information) with a computable one (the largest eigenvalue of the Gram matrix). However, the mathematical bridge is not viable. At every critical juncture, the argument either assumes what it needs to prove, relies on conditions that the target application manifestly violates, or substitutes heuristics for theorems. I detail the fatal flaws below.

1. The Information-Spectral Bridge is Fundamentally Misconceived

The core claim is that $\hat{\delta} = \max(0, (\lambda_1 - \text{MP}_+)/C)$ serves as a proxy for $\delta = I(\phi; S)$. This conflates two quantities that measure

fundamentally different things.

Mutual information $I(\phi; S)$ measures the reduction in uncertainty about the state labels S given the features ϕ . It is a functional of the full joint distribution $P(\phi, S)$ and depends on every dimension of ϕ , every interaction between dimensions, and the detailed geometry of the conditional distributions.

The largest eigenvalue λ_1 measures the maximum variance direction of the feature distribution. It is a one-dimensional summary statistic of the marginal distribution $P(\phi)$. It does not, on its own, tell you whether that variance is aligned with the state labels.

Counterexample 1A: Large λ_1 , small $I(\phi; S)$. Let S be a binary label independent of all features. Construct ϕ such that the first coordinate is $Z \sim N(0, 1000)$ and coordinates $2\dots d$ are i.i.d. $N(0,1)$. The population covariance is $\Sigma = \text{diag}(1000, 1, \dots, 1)$. The largest eigenvalue is 1000, far exceeding the MP bulk. The BBP test declares “strong features.” Yet $I(\phi; S) = 0$ because ϕ and S are independent. The spectral proxy gives a large positive $\hat{\Delta}$ while the true Δ is zero. This is not a pathology — it is the expected behavior whenever the dominant variance direction is not the state-separating direction. The author implicitly assumes that the top eigenvector of the covariance matrix aligns with the state separation, which is exactly the assumption that needs to be tested, not assumed.

Counterexample 1B: Small λ_1 , large $I(\phi; S)$. Let S be a 50-state categorical variable. For each state k , construct the state mean μ_k as a vector in $\mathbb{R}^{1000}$ where exactly 20 randomly chosen coordinates are set to 0.5 and the rest are 0. The state means differ in their support patterns, not in their overall norm. The between-state scatter B has rank at most 49, but each eigenvalue of B is small (approximately $0.5^2 * 20/1000 = 0.005$ per

coordinate, times a multiplicity factor). The largest eigenvalue of B/σ^2 may be well below $\sqrt{\gamma}$ even for modest γ . Yet the mutual information between ϕ and S is substantial: the 20 non-zero coordinates provide nearly perfect state discrimination because each state’s support pattern is unique. The spectral proxy returns $\hat{\delta} = 0$. The true δ is approximately $\log_2(50) = 5.6$ bits. False negative: catastrophic.

The defense in Section 12.1 (“an eigenvalue spike can only appear if there genuinely is between-state variation”) is incorrect, as Counterexample 1A shows. A high-variance feature independent of S produces a spike just as effectively as a state-separating feature. The spectral proxy cannot distinguish between “signal-relevant variance” and “nuisance variance.”

2. The “Monotonic Relationship” Claim is Unsupported

Section 3.4 states: “The effective signal strength θ and the mutual information δ are **monotonically related** under the isotropic Gaussian mixture.”

The supporting evidence is:

1. An **upper bound**: $\delta \leq (K-1) * \theta / 2$ (Section 3.2, item 1).
2. A **lower bound** for the binary case: $\delta \geq \theta/8$ (Section 3.2, item 2).
3. A **different lower bound** for equi-separated means: $\delta \geq (1/2) \log(1 + \theta/(K-1))$ (Section 3.3).

These bounds do not establish monotonicity. They establish that δ lives

in the interval $[\theta/8, (K-1)\theta/2]$ for binary states, which spans a factor of $4(K-1)$. Monotonicity would require proving that for any two parameter configurations $(\theta_1, \text{mean geometry}_1)$ and $(\theta_2, \text{mean geometry}_2)$, if $\theta_1 > \theta_2$ then $\delta_1 > \delta_2$. The provided bounds are too loose to guarantee this.

A concrete failure of monotonicity under the author’s own model.

Consider two configurations, both binary Gaussian mixtures with isotropic noise:

- Configuration A: $\mu_1 = [\theta^{1/2}, 0, \dots, 0]$, $\mu_2 = [-\theta^{1/2}, 0, \dots, 0]$. Then $\theta = 4\theta$ (since $\theta_{\text{def}} = \|\mu_1 - \mu_2\|^2 / (4\sigma^2)$ from Section 8.1) $= (2\theta^{1/2})^2 / 4 = \theta$. The mutual information is approximately $(1/2)\log(1 + \theta/d)$ for large d , which is tiny for small θ/d .
- Configuration B: $\mu_1 = [0, \dots, 0]$, $\mu_2 = [\epsilon, \epsilon, \dots, \epsilon]$ with $d/2$ non-zero entries. Then $\theta = \|\mu_1 - \mu_2\|^2 / (4\sigma^2) = (d/2)\epsilon^2 / (4\sigma^2) = d\epsilon^2 / (8\sigma^2)$. For $\epsilon = \sqrt{8\sigma^2 \theta / d}$, we get the same θ value. But the mutual information is approximately $(d/2)(\epsilon^2 / (2\sigma^2)) = (d/2)(4\theta/d) = 2\theta$, much larger than Configuration A.

Same θ , different δ s — monotonicity is false even within the isotropic Gaussian mixture model. The relationship between θ and δ depends not just on θ but on the **geometry** of the state means, which the spectral proxy discards.

The sentence “The mapping is not one-to-one (θ can be large while δ saturates at $\log K$), but it is a valid sufficient condition” is a hedge that admits the mapping is not injective, which is exactly the problem. A sufficient condition that says “if the spectral spike is large, features might be strong,

or then again they might not be” is not a theorem — it is a heuristic.

3. The Gaussian Mixture Assumption is Fatal for SCX’s Actual Features

The entire derivation assumes:

- $\phi(x) \mid S = k \sim N(\mu_k, \sigma^2 I_d)$ (isotropic Gaussian within-state)
- The covariance is exactly $\sigma^2 I_d + \text{low-rank } B$ (spiked structure)
- The entries of Φ are i.i.d. (for the MP law and TW fluctuations)

SCX’s actual features violate every one of these assumptions:

1. **ACE descriptors** for ALN (the primary SCX use case) are body-order expansions of atomic neighbor densities computed from radial basis functions and spherical harmonics. These are:
 - NOT Gaussian: they are sums of polynomial terms evaluated at neighbor distances, with nonlinear cutoff functions. Their marginal distributions have heavy tails and non-zero skewness.
 - NOT isotropic: different ACE channels (body-order, radial channel, angular channel) have vastly different variances, ranges, and correlation structures. The within-state covariance is highly anisotropic.
 - NOT i.i.d. across samples: atomic configurations that share atoms (in supercell calculations with periodic boundary conditions) produce correlated descriptors. The i.i.d. assumption is violated structurally.

2. **CNN latents** (CIFAR-10, DermaMNIST) are outputs of trained neural networks with ReLU activations and max-pooling. These are:
 - NOT Gaussian: the distribution of CNN activations in the penultimate layer is known to be non-Gaussian — it often exhibits a “cone” structure (samples from the same class lie on a low-dimensional manifold embedded in a high-dimensional space).
 - NOT isotropic: CNNs produce highly anisotropic representations. The singular value distribution of CNN feature matrices is known to have a slow decay, not the sharp spectral gap assumed by the spiked model.
 - NOT i.i.d.: the features are deterministic functions of the inputs, and the model architecture induces correlations.
3. **Tabular features** (generally) have arbitrary marginal distributions, missing values, mixed continuous/categorical types. The Gaussian assumption is not approximately true.

The author acknowledges these violations in Section 13.2 but offers only hand-wavy mitigations: “pre-whiten,” “permutation test,” “robust test.” None of these are part of the theoretical bridge. The “pre-whiten” suggestion requires estimating the within-state covariance, which is what we’re trying to avoid computing (it requires knowing the state partition). The “permutation test” suggestion replaces the parametric spectral test with a nonparametric one, which abandons the Tracy-Widom theory entirely. The “robust test” suggestion (citing Berthet & Rigollet, 2013) addresses a different problem (sparse PCA, not general non-Gaussian features).

The core issue: Under misspecification, the MP bulk edge MP_+ is no longer correct. For anisotropic within-state covariance, the limiting spectral distribution of the Gram matrix is a free convolution of the MP law with the population eigenvalue distribution of Σ_W . The upper edge can

be arbitrarily far from $\sigma^2 (1+\sqrt{\gamma})^2$. The BBP threshold $\sqrt{\gamma}$ depends on the isotropy assumption. Without isotropy, both the null distribution and the phase transition location are unknown.

4. The Tracy-Widom Test is Invalid for SCX Features

The test statistic uses TW_1 critical values derived for the largest eigenvalue of a sample covariance matrix whose entries are i.i.d. with finite 4th moment.

SCX features violate the i.i.d. entry assumption in multiple ways:

1. **Row dependence:** For ACE descriptors computed from atomic configurations, samples share atoms (via periodic boundary conditions, sliding windows, or augmentation). The rows of Φ are not independent. With dependence, the fluctuations of λ_1 can be much larger than TW_1 predicts, producing anti-conservative tests (more false positives than the nominal level).
2. **Column dependence:** Feature dimensions are not independent. ACE descriptor channels are highly correlated (nearby radial channels, angular harmonics). CNN latent dimensions have complex correlation structures induced by the network architecture. The TW_1 result assumes independence across both rows and columns.
3. **Fourth moment condition:** The TW_1 universality class requires finite 4th moment of the entry distribution. ACE descriptors involve products of radial basis functions that can produce large values near the cutoff, potentially violating moment conditions. CNN latents with ReLU activations have heavy-tailed behavior in the deeper layers (the

“edge of chaos” phenomenon).

4. **Dimensionality regime:** The Tracy-Widom centering and scaling formulas (Section 5.2) assume $d/N \rightarrow \gamma$ in $(0, \infty)$. For ALN with $d \sim 100$ -200 and $N = 534$, $\gamma \sim 0.2$ -0.4, which is moderate. However, the convergence rate to TW_1 is known to be slow — $O(N^{\{-2/3\}})$ in the bulk, but the edge may converge slower, especially for non-Gaussian data. The finite- N approximation errors are unquantified.

The power analysis in Section 5.4 computes “detectable gaps” for the three datasets, assuming TW_1 applies perfectly. These numbers (epsilon_detect of 0.12-0.27) are optimistic precisely because they ignore the excess variance from non-i.i.d. features.

5. The Calibration Constant C is a Free Parameter

The treatment of C is the weakest part of the paper, revealing that the “theorem” is not a theorem at all.

Step 1: Theoretical derivation (Section 8.1). The author derives $C = 2 * \sigma^2 * (\theta - \sqrt{\gamma})^2 / (\theta * (1 + \theta))$. But this depends on the unknown θ , which is what we’re trying to infer from lambda_1. Circular.

Step 2: Self-consistent estimation (Section 8.2). The author proposes inverting the BBP formula to estimate θ from lambda_1, then plugging θ into C. But this is algebraically redundant: if you already have θ from lambda_1 via the BBP inverse formula, you don’t need the spectral proxy at all — you can compute delta directly from θ (under the Gaussian

model). The “self-consistent estimation” reveals that the proxy is just a complicated change of variables, not an independent quantity.

Step 3: Practical default $C = 2$ (Section 8.3). The justification is: “From the binary Gaussian mixture: $\delta = \theta/2$ and $\Delta_\lambda = \sigma^2 \theta$. So $C = 2 \sigma^2$. After normalizing σ^2 to 1, $C = 2$.” This is valid only in the regime where: - γ is negligibly small (so $\sqrt{\gamma} \rightarrow 0$ and the full formula collapses) - $\theta \gg \sqrt{\gamma}$ (so the “ $(\theta - \sqrt{\gamma})^2/(1+\theta)$ ” term simplifies to θ) - $K = 2$ (binary states only) - The feature dimension d is large enough that $\delta = \theta/2$ (which requires $d \gg \theta$, so the Gaussian mixture mutual information formula linearizes)

All four conditions fail for typical SCX data. For DermaMNIST with $\gamma = 0.0064$, $\sqrt{\gamma} = 0.08$ — not negligible relative to small θ values. For $K = 7$ classes (DermaMNIST), the single-spike proxy misses 6 dimensions of state separation.

What is the worst-case error if $C=2$ but the true value is $C=5$? Then $\hat{\delta} = (\lambda_1 - \mu_+)/2$ instead of $(\lambda_1 - \mu_+)/5$, overestimating δ by a factor of 2.5. The plug-in F1 bound becomes: - $F1_{SCX} \leq F1_{base} + C_F * \sqrt{2 * \delta_{hat_true} * 2.5} - F1_{base} + 1.58 * C_F * \sqrt{\delta_{hat_true}}$

This is 58% larger than the correct bound, meaning the diagnostic is optimistic — it says “SCX might work” when the true bound says it won’t. False positives in the non-conservative direction.

If $C=2$ but the true value is $C=0.5$? Then $\hat{\delta}$ underestimates δ by a factor of 4, making the diagnostic overly conservative — it rejects SCX when SCX would work. The false negative rate is uncontrolled.

The document provides **no** bound on $|C_true - C_used|$, **no** sensitivity analysis, and **no** error propagation. This is not a theorem — it is an ad-hoc calibration.

6. The “Sufficient but Not Necessary” Defense Acknowledges the Fatal Flaw

Section 12.2 admits: “If λ_1 is below the MP edge, it does NOT guarantee SCX will fail.” This is not a minor caveat — it is the core failure mode of the diagnostic.

How conservative is the test? This depends on the dataset and requires specifying: - The number of states K - The geometry of the state means (are they collinear, orthogonal, or uniformly distributed?) - The noise level σ^2 - The aspect ratio γ

For the worst-case scenario of $K = 50$ states with means uniformly distributed on a sphere in $\mathbb{R}^d$, the leading eigenvalue of B/σ^2 is approximately $R^2/(K * \sigma^2) * (1 + (d-1)/K) \approx R^2/(K\sigma^2)$ for large d . The mutual information is approximately $(R^2/(2\sigma^2) * \text{tr}(B) / \sigma^2) / (R^2/(2\sigma^2) * K * (R^2/(K\sigma^2))) = R^4/(2\sigma^4)$. For $R^2/\sigma^2 = 10$ (moderate signal), $K = 50$, $\gamma = 0.1$, $d = 100$:

- $\theta_1 = 10/50 = 0.2$, $\sqrt{\gamma} = 0.316 \rightarrow \theta_1 < \sqrt{\gamma}$, subcritical, $\hat{\Delta} = 0$.
- True $\Delta = (10^2)/2 = 50$ nats (for the spherical configuration, approximately). This is enormous.

This is a false negative rate of 100% for this configuration. The diagnostic says “weak features, don’t use SCX” for a scenario where the features are

actually highly informative.

Quantification attempt. Let $B \sim \text{Wishart}(K-1, \text{diag}(\theta_1, \dots, \theta_{K-1}))$. The largest eigenvalue of B follows a Tracy-Widom distribution around θ_1 . The condition for non-detection is:

$$\theta_1 \leq \sqrt{\gamma} = \sqrt{d/N}$$

For K large with approximately equal separation in all directions, θ_1 (mean separation per direction) $\approx (d/K)$. If $K > d/\sqrt{\gamma}$, then $\theta_1 < \sqrt{\gamma}$ even when the total signal is strong. The false negative probability is approximately:

$$P(\text{false negative} \mid \text{total signal strong}) \approx 1 \text{ for } K > d / \sqrt{d/N} = \sqrt{N \cdot d}$$

For SCX regimes: $N = 500$, $d = 100 \rightarrow \sqrt{500 \cdot 100} \approx 224$. If $K > 224$ (which is unrealistic for SCX). For more typical $K = 10$, the false negative condition is less extreme but still possible when means are nearly collinear with small separation along each direction.

The document does not provide this calculation, presumably because it exposes the severity of the conservatism problem.

7. The σ^2 Estimation is Biased and the Bias is Not Corrected

$MP_+ = \sigma^2 (1 + \sqrt{\gamma})^2$ requires σ^2 . The author estimates σ^2 from trailing eigenvalues (Section 5.3, step 3): $\hat{\sigma}^2 = \text{mean of bulk eigenvalues after discarding spikes}$.

Problem 1: Signal contamination. Under the spiked model, the total sample variance is:

$$\text{tr}(\hat{\text{Sigma}}) = (1/N) * \text{tr}(\text{Phi} \text{Phi}^T) = \sigma^2 * d/N + (1/N) * \text{tr}(\text{Phi_mean} \text{Phi_mean}^T)$$

The sample bulk eigenvalues are not exactly σ^2 — they are σ^2 * (eigenvalues of a $d-N(0,I)$ Wishart matrix) perturbed by the spikes. Even after removing the top r eigenvalues, the remaining bulk eigenvalues have expectation:

$$E[\lambda_{\{r+1\}}] = \sigma^2 * (1 + \gamma * \eta_j) \text{ where } \eta_j \text{ depends on the spike strengths}$$

The bias is $E[\hat{\sigma}^2] - \sigma^2 = \sigma^2 * \gamma * (1/N) * \sum_{j=r+1}^d \eta_j$, which is positive. This inflates MP_+ , making the test conservative — but the **amount** of conservatism is dataset-dependent and not quantified.

Problem 2: The “elbow” method for counting spikes is unreliable. Algorithm step 2 uses a threshold rule ($\lambda_j > 2 * \text{median}(\lambda)$) to count spikes. This is a heuristic with no theoretical justification. For weak spikes near the detection threshold, the count r_{est} is wrong, contaminating the bulk estimate.

Problem 3: The bulk trimming is arbitrary. The algorithm discards “ $r_{\text{est}} + 10$ ” eigenvalues. The “+10” is a guess. If r_{est} is incorrect by more than 10 (plausible for complex spectral structure), the bulk estimate is contaminated.

Problem 4: The alternative estimator (median / $MP_{\text{median_factor}}$) has its own issues. The median of the MP distribution is a known function

of γ , but for finite N , the sample median deviates from the population median, and this deviation is comparable to the bulk fluctuation scale.

The combined effect: for AlN with $d = 100$, $N = 534$, $\gamma = 0.19$, if the true $\sigma^2 = 1$ but estimate yields $\hat{\sigma}^2 = 1.15$ (15% relative bias, plausible with signal contamination), then MP_+ is inflated by 15%, making the critical value $(1 + \sqrt{0.19})^2 = 1.64$ become 1.89. This 15% increase in the threshold can mask a moderately strong spike ($\theta = 0.5$, $\lambda_1 = 1.8$) as subcritical.

8. The Comparison to Lei & Zhu (2018) Reveals No Mathematical Contribution

Lei & Zhu (2018) already proved:

1. Under a spiked covariance model (their Assumption 2.1), spectral k-means achieves a misclassification rate of $O(\exp(-c * \text{eigengap}^2 * N))$.
2. The detectability condition is that the leading eigenvalue of the signal matrix exceeds the noise level, which is exactly the BBP condition.
3. The misclassification rate depends on the eigengap, which is proportional to the spectral excess ($\lambda_1 - \lambda_2$).

What this document adds beyond Lei & Zhu:

Claimed novelty (Section 10.2): “SCX does spectral clustering on the GRAM matrix ($N \times N$) while Lei & Zhu work with the feature covariance matrix ($d \times d$).” This is mathematically vacuous. The non-zero eigenvalues of $\Phi \Phi^T$ ($N \times N$) and $\Phi^T \Phi$ ($d \times d$) are identical up to scaling. Working with the Gram matrix is a computational convenience, not a mathematical difference.

Claimed novelty (Section 10.1, item “Our addition”): “Connecting the spectral spike to an information-theoretic quantity (mutual information).” But this connection is: - Not proved (it relies on the heuristic calibration C, as shown in Point 5) - Not needed (Lei & Zhu’s eigengap condition is already a valid sufficient condition for clustering success) - Not robust to model misspecification (as shown in Point 3)

Claimed novelty (Section 11.1): “The specific connection between the BBP eigenvalue phase transition and SCX’s mutual information bound is novel.” Novel and false are not mutually exclusive. The connection does not survive mathematical scrutiny.

The honest assessment comes from the author’s own novelty table (Section 11.3): - “BBP phase transition”: 0/5 (known) - “k-means = PCA on Gram”: 0/5 (known) - “Spiked covariance model”: 0/5 (known) - “Spectral clustering consistency”: 0/5 (known) - “Tracy-Widom test for spikes”: 1/5 (standard) - “Mapping theta $\rightarrow$ delta”: 3/5 (but this is the part I’ve shown to be unsupported) - “Calibration C estimation”: 3/5 (but this is circular, as shown) - “Plug-in diagnostic for Thm 2”: 4/5 (but it inherits all the flaws of the upstream connections) - “Multi-spike proxy”: 2/5 (natural extension) - “Spectral proxy algorithm”: 3/5 (combines known pieces)

The weighted novelty is even lower than the author’s estimate, because the highest-scoring items (“Mapping theta $\rightarrow$ delta” and “Calibration C”) are the ones with the most serious mathematical defects. When these collapse, the “Plug-in diagnostic for Thm 2” collapses with them.

Additional Mathematical Issues

9. The “Proof Sketch” is Not a Proof

Section 4.2 calls itself a “proof sketch” but contains no proof. It consists of:

- Part (a): “follows directly from the BBP phase transition” — correct, but this is just restating BBP, not connecting to SCX.
- Part (b): “requires connecting the excess eigenvalue to the mutual information δ ” — this is the statement of what needs to be proved, not a proof. The subsequent bullet points do not constitute a proof:

- Bullet 1: When $\theta \leq \sqrt{\gamma}$, $\delta_{\text{hat}} = 0$ and “ δ can be arbitrarily small or zero in this regime.” This is false. δ can be non-zero even when $\theta \leq \sqrt{\gamma}$ (Counterexample 1B).
- Bullet 2: The excess eigenvalue formula is correct, but the inequality “ $\delta_{\text{hat}} \leq C * \delta$ ” is not derived — it’s asserted with the qualifier “for some C depending on γ and the state geometry.”
- Bullet 3: The calibration constant is chosen “so that δ_{hat} is conservative.” But this is circular: C is defined as $(\lambda_1 - \mu_+) / \delta_0$, where δ_0 is a “calibrated reference point.” The choice of δ_0 determines the conservatism, and there is no principled way to choose it without knowing the true δ .

- Part (c): “follows from applying Theorem 2’s proof with δ_{hat} in place of δ , plus an extra error term.” This assumes $\delta_{\text{hat}} \leq \delta$, which is exactly what Part (b) was supposed to prove but did not.
- Part (d): “relies on Tracy-Widom fluctuation theory.” The convergence rate $O(N^{-2/3})$ is for the TW centering and scaling of λ_1 , not for δ_{hat} . The error in δ_{hat} is propagated through the calibration constant C , which has its own error that is $O(1)$ — dominating

the TW error.

10. Part (b) of the Theorem is Internally Contradictory

The theorem states two inequality claims: - (b1) $\delta_{\text{hat}} \leq \delta + o(1)$ [upper bound: proxy is conservative] - (b2) $\delta_{\text{hat}} \geq c(\theta) * \delta + o(1)$ [lower bound: proxy is proportional]

For (b1) to hold when $\theta \leq \sqrt{\gamma}$, we need $\delta_{\text{hat}} = 0 \leq \delta + o(1)$, i.e., $\delta \geq 0$. This is true (mutual information is non-negative). But for (b2) to hold when $\theta > \sqrt{\gamma}$, we need δ_{hat} to be a non-trivial fraction of δ . The constant $c(\theta)$ is not specified. For θ just above $\sqrt{\gamma}$, the spectral excess is $O((\theta - \sqrt{\gamma})^2)$, while δ is $O(\theta)$. Therefore $c(\theta) = O((\theta - \sqrt{\gamma})^2 / \theta) \rightarrow 0$ as $\theta \rightarrow \sqrt{\gamma}^+$. The “lower bound” becomes vanishingly small exactly when we need it most — near the detection boundary.

11. The Plug-in to Theorem 2 (Section 6) is Circular

The bound is: $F1_{\text{SCX}} \leq F1_{\text{base}} + C_F * \sqrt{2 * \delta_{\text{hat}}} + \text{error}$

The author claims this is “valid with probability $1 - o(1)$ (asymptotically).” But: - Theorem 2’s original bound uses $\delta = I(\phi; S)$ and is a deterministic inequality valid for ALL distributions. - The proxy version replaces δ with δ_{hat} , which is a random variable depending on the sample. - The claimed validity requires $\delta_{\text{hat}} \leq \delta + o(1)$ with high probability (Part b1), which is NOT proved. - The error term $\epsilon_N = O(N^{-2/3})$ from TW fluctuations is dwarfed by the calibration error $O(1)$ from C .

The statement “The bound remains valid” assumes the conclusion.

12. The Validation Predictions (Section 9) are Tautological

The three datasets are chosen to match the expected outcomes: - AIN (strong SCX improvement) -> “strong features” - CIFAR-10 (moderate SCX improvement) -> “strong features” (safe prediction because gamma is tiny) - DermaMNIST (weak SCX improvement) -> “weak features” (safe prediction because SCX doesn’t work)

The document proposes comparing $\hat{\Delta}$ to “observed SCX F1 improvement” but does not compute either quantity for any dataset. The “cross-dataset behavior” table (Section 9.5) has no actual numbers for λ_1 or $\hat{\Delta}$ — they’re all placeholders (“» MP”, “> 5.0”, “high”). The table’s “ $\hat{\Delta}$ ” column is filled with qualitative labels, not numbers.

This is not validation. It is storytelling.

13. The “Many Weak Signals” Regime (Section 13.3) is Dismissed Too Quickly

The author acknowledges that for K states with small but balanced separation, all eigenvalues can be subcritical while the total mutual information is large. The proposed mitigation is the “multi-spike proxy” (Section 8.4), which sums excess eigenvalues.

But the multi-spike proxy inherits the same problem: if each $\lambda_j - \text{MP}_+$ is negative or zero (all eigenvalues within the bulk), the sum is zero. The “many weak signals” regime is not addressed by summing — it requires

detecting structure BELOW the MP edge, which the spiked model explicitly cannot do.

The author’s worst-case analysis (Section 13.3) uses the example of $K \geq d$ with means on a sphere. The analysis shows that each eigenvalue is $O(R^2/K)$. For $K \geq d/\sqrt{\gamma}$, all eigenvalues are subcritical. The author admits: “The spectral proxy (single-spike) would set $\hat{\delta} = 0$, giving a false negative.” The mitigation offered is the multi-spike proxy, but the multi-spike proxy fails in the same regime because all eigenvalues are subcritical.

This is not a niche case. For SCX’s actual use case of material science with ACE descriptors, the “states” are local atomic environments: perfect crystal, various defect types, surface sites, grain boundaries. There can be dozens of distinct states. The state means lie on a low-dimensional manifold embedded in a high-dimensional descriptor space. Individual eigenvalues of B may well be small. The spectral proxy would diagnose “weak features” even when the descriptors are highly informative.

Conclusion

The document claims to build a mathematical bridge between the BBP phase transition and SCX’s mutual information bound. The bridge is not built. What exists is:

1. An asymptotic relationship between the largest eigenvalue and the effective signal strength θ (from BBP), which is correct but is not new and does not involve mutual information.

2. A set of inequalities relating θ and δ under the isotropic Gaussian mixture model, which are too loose to establish the claimed monotonicity and fail for multiple-signal geometries.
3. A calibration constant C that is either circular (depends on the unknown θ), heuristic (default $C=2$), or unvalidated (no error bounds).
4. A Tracy-Widom test that is invalid for SCX’s non-i.i.d., non-Gaussian, anisotropic features.
5. A plug-in diagnostic that inherits all the defects above and adds no new mathematical guarantees.

The author’s own novelty assessment (2.5/5) and publication potential assessment (“weak” for a full paper) are more honest than the theorem statements suggest. The spectral proxy may have practical value as a heuristic screening tool, but the document’s framing as a mathematical theorem with Parts (a)-(d), Corollaries, and rigorous asymptotic claims is misleading. The “theorem” is not a theorem — it is a collection of insights from random matrix theory assembled around a heuristic proxy, held together by unvalidated assumptions and circular reasoning.

Recommendation: Reject. Do not resubmit in current form.

End of review.